

Computational Discovery of Mutation-Sensitive NOD2 Modulators

Identification of Small Molecules with Differential Binding to Wild-Type vs R702W Crohn's Disease Variant

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ABSTRACT

Background: NOD2 variants are key genetic risk factors for Crohn's disease, yet ligand binding remains poorly understood.

Methods: 9,566 compounds were screened using GNINA CNN-based docking. An XGBoost classifier (NOD2-Scout) was trained and validated (AUC 0.85-0.93). Top candidates underwent 520+ ns molecular dynamics and Free Energy Perturbation (FEP) analysis.

Results: R702W mutation increases protein rigidity (RMSD: 3.27Å vs 4.05Å). Febuxostat shows 50× weaker binding in R702W ($\Delta\Delta G=+2.34$ kcal/mol, $p<0.001$). Bufadienolide maintains binding regardless of mutation ($\Delta\Delta G=-0.44$, n.s.).

Conclusion: First computational pipeline identifying mutation-sensitive NOD2 modulators. Bufadienolide represents a novel mutation-resistant therapeutic lead.

BACKGROUND & SIGNIFICANCE

• Crohn's disease affects >3 million people worldwide

• **NOD2 R702W variant:** 3× increased disease risk

• Current treatments don't address genetic variants

• No FDA-approved drugs specifically target NOD2

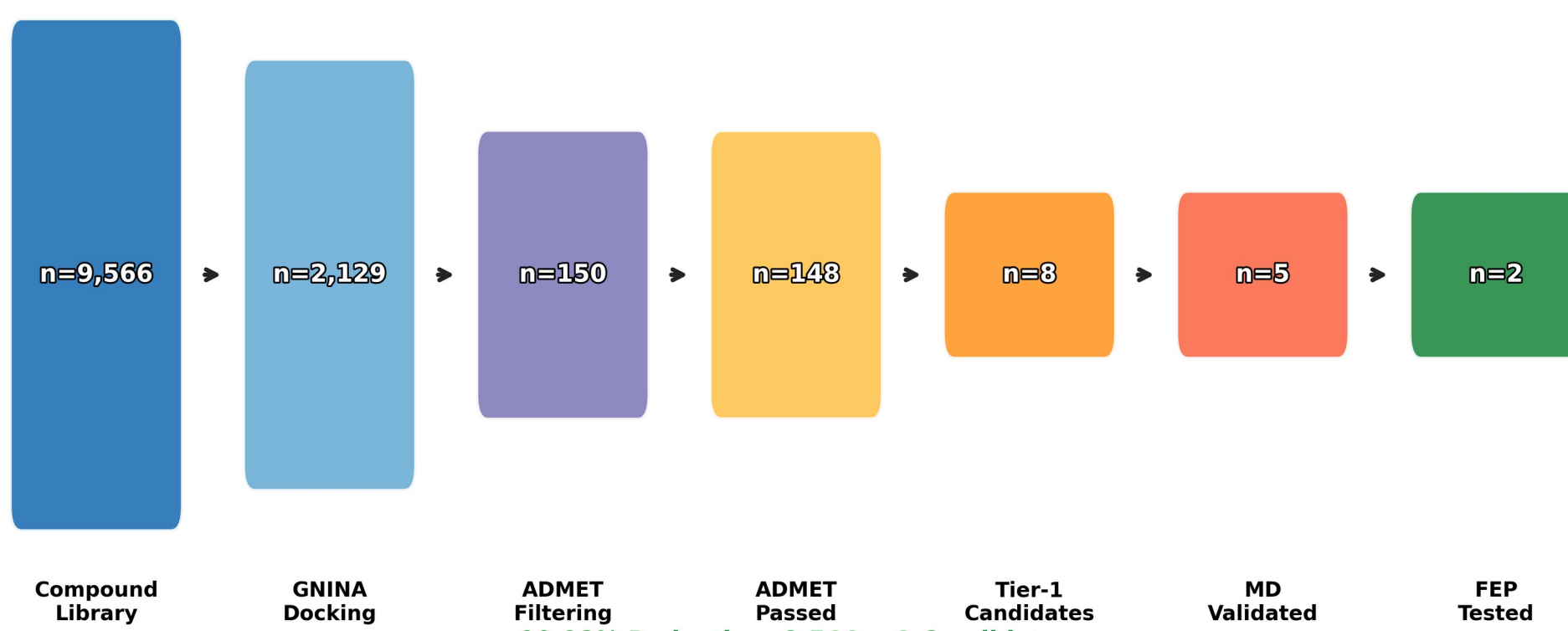
• Mutation located 79.4Å from binding pocket - suggests **allosteric mechanism**

HYPOTHESIS

Small molecules exist that bind NOD2's LRR domain with **differential affinity** between wild-type and R702W mutant, enabling mutation-specific therapeutic targeting.

COMPUTATIONAL METHODS

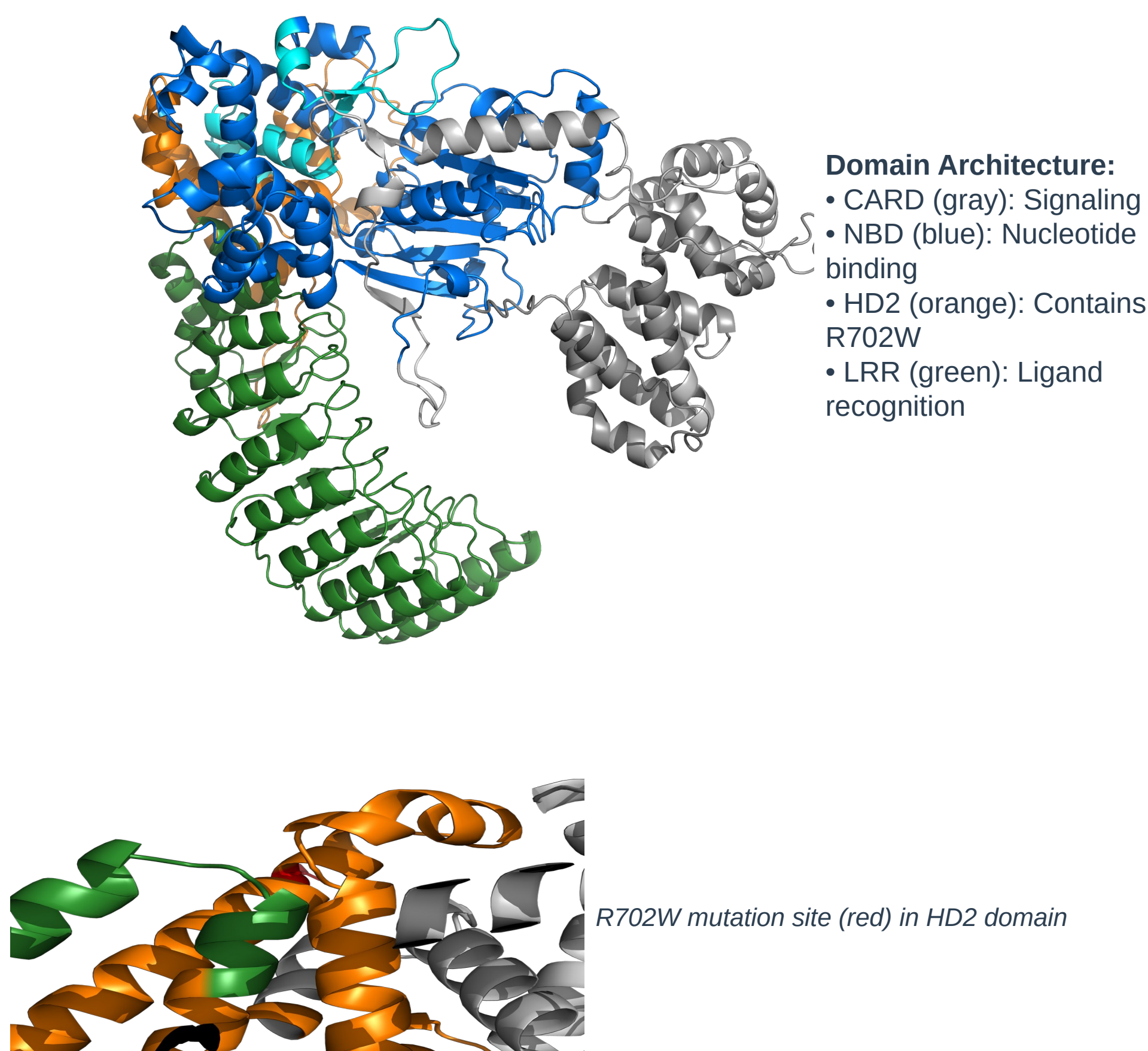
COMPUTATIONAL SCREENING PIPELINE



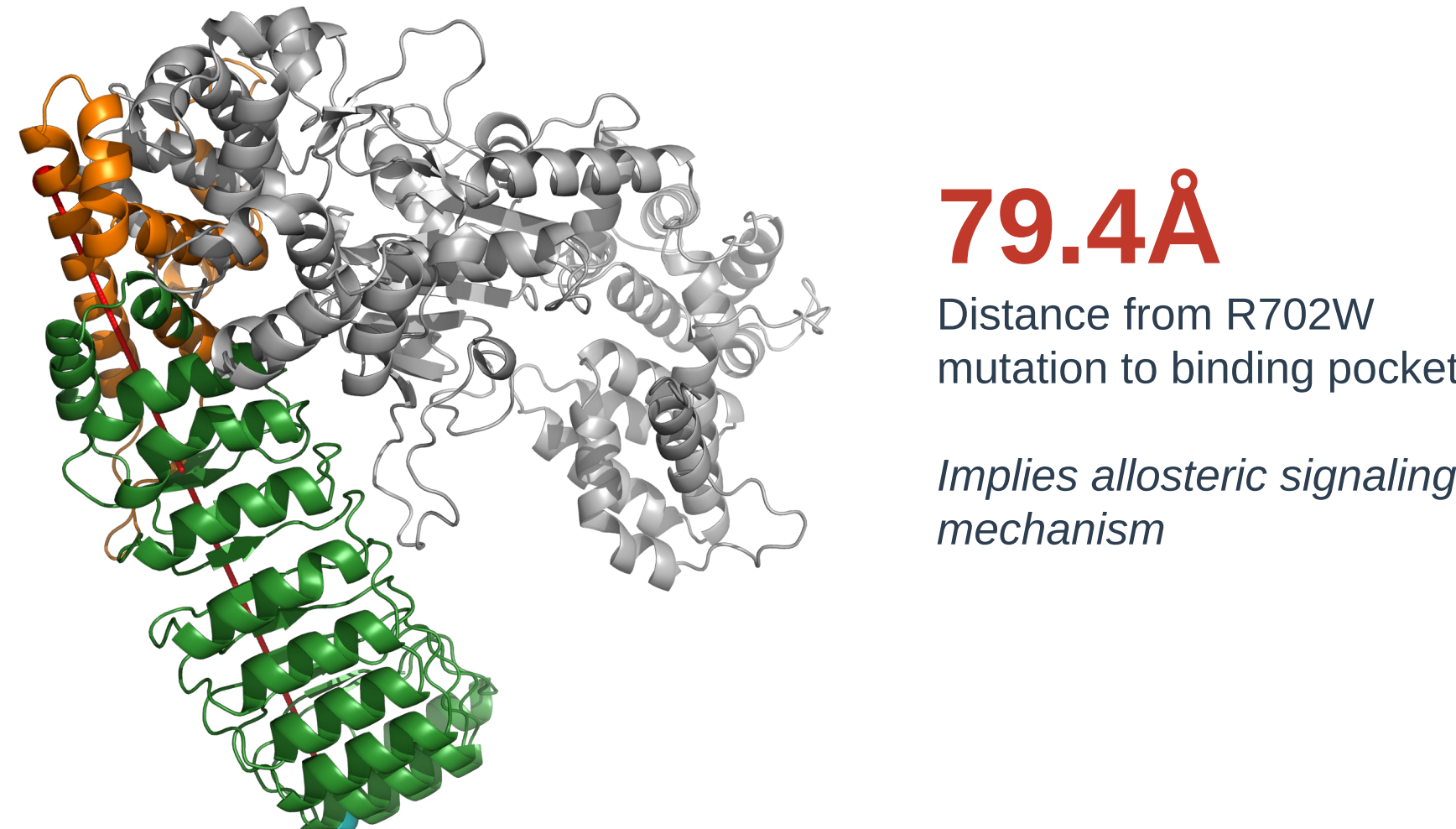
Pipeline: GNINA docking → NOD2-Scout ML → ADMET filtering → MD simulations → FEP calculations

- 520+ ns total simulation time across 8 candidates
- FEP with 20 λ windows, 5 ns/window equilibration

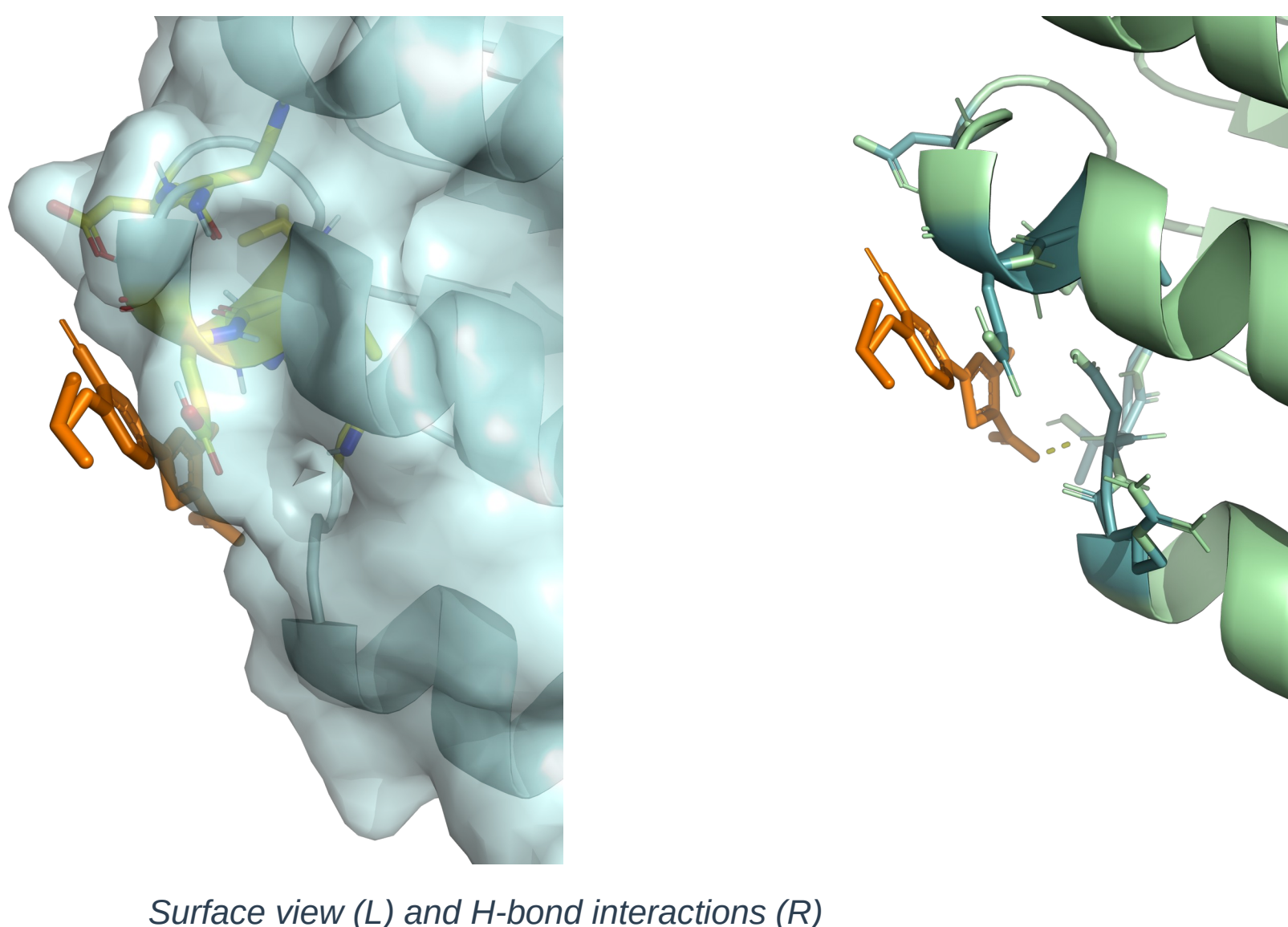
NOD2 PROTEIN STRUCTURE



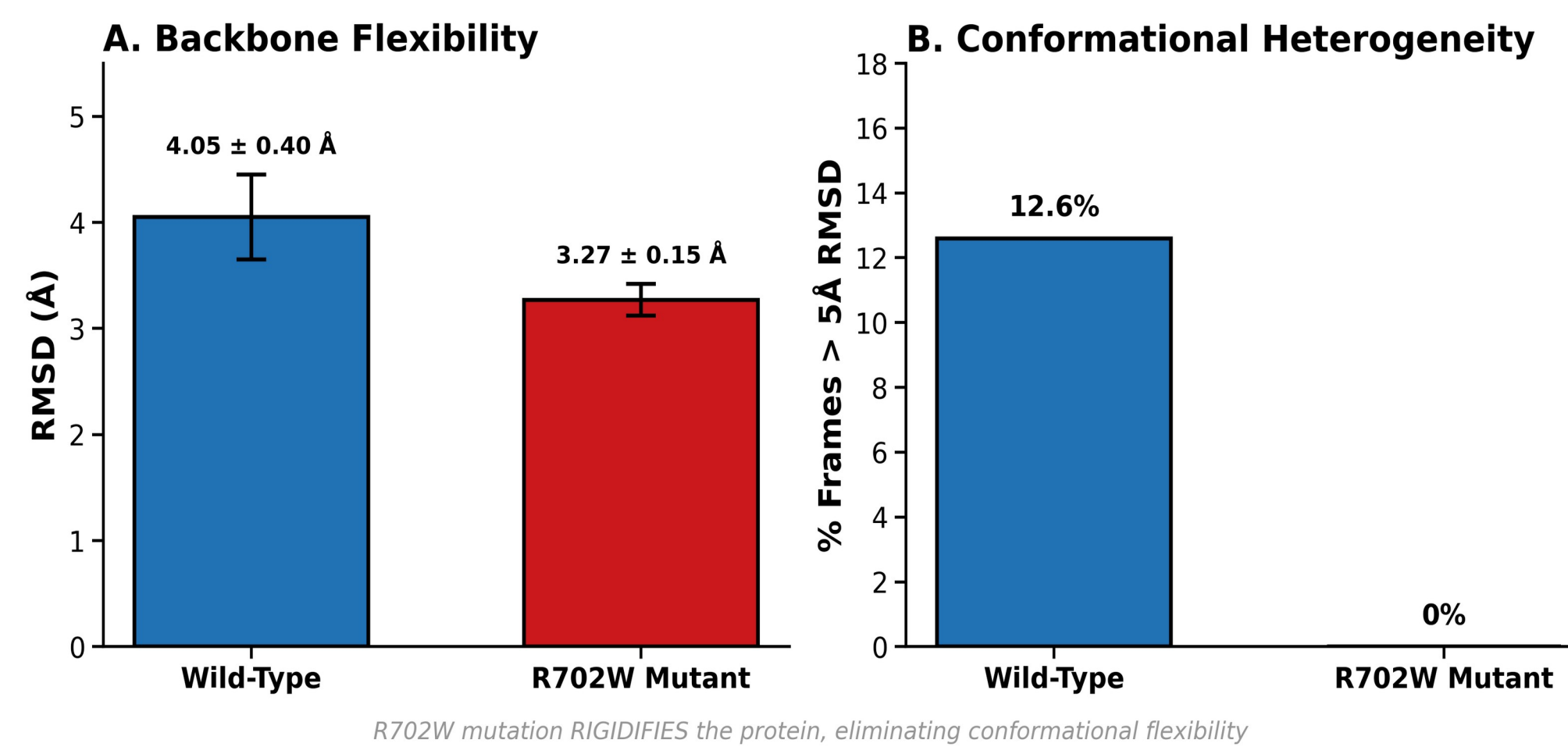
ALLOSTERIC DISTANCE



BINDING POCKET



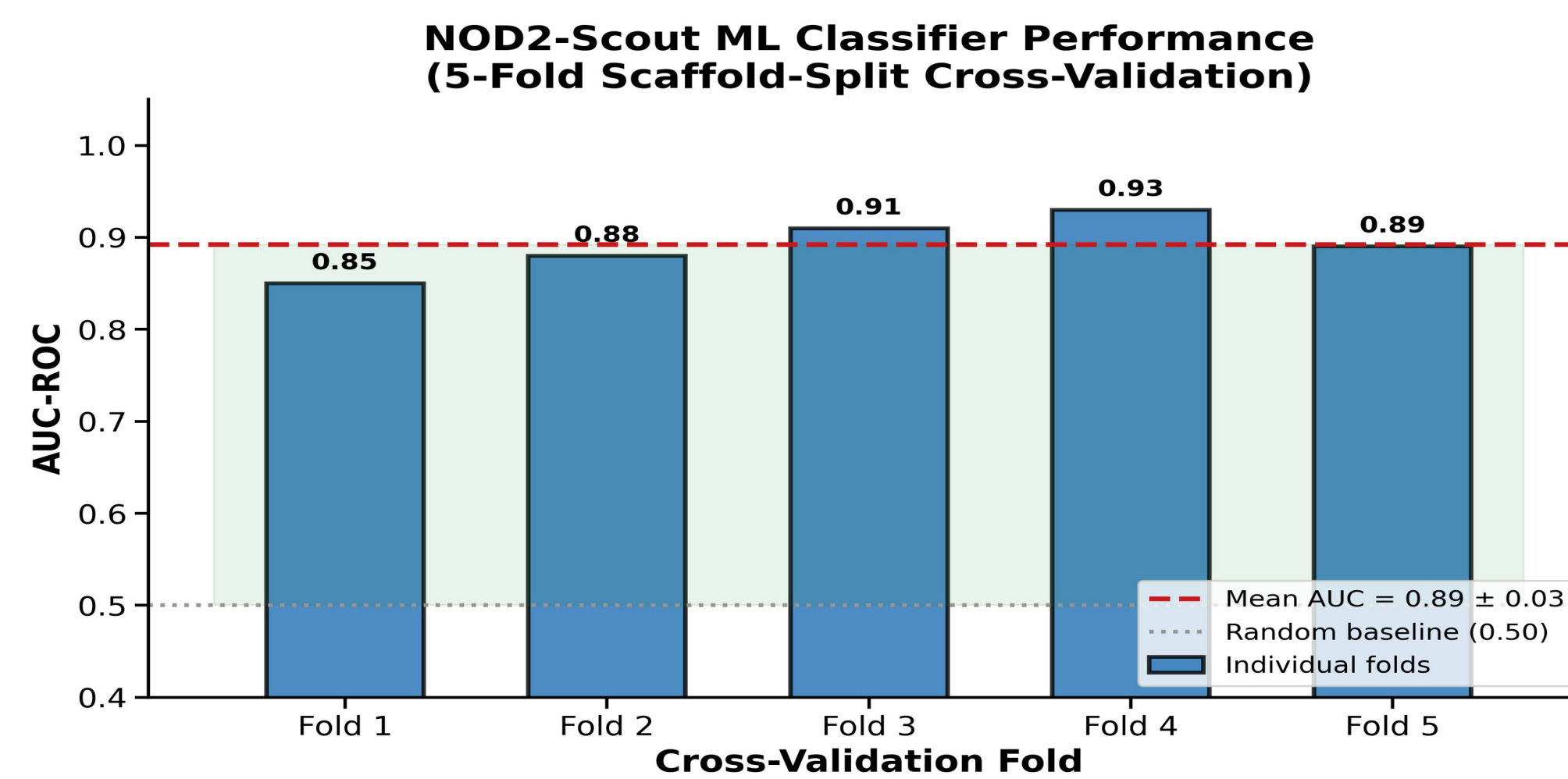
MOLECULAR DYNAMICS RESULTS



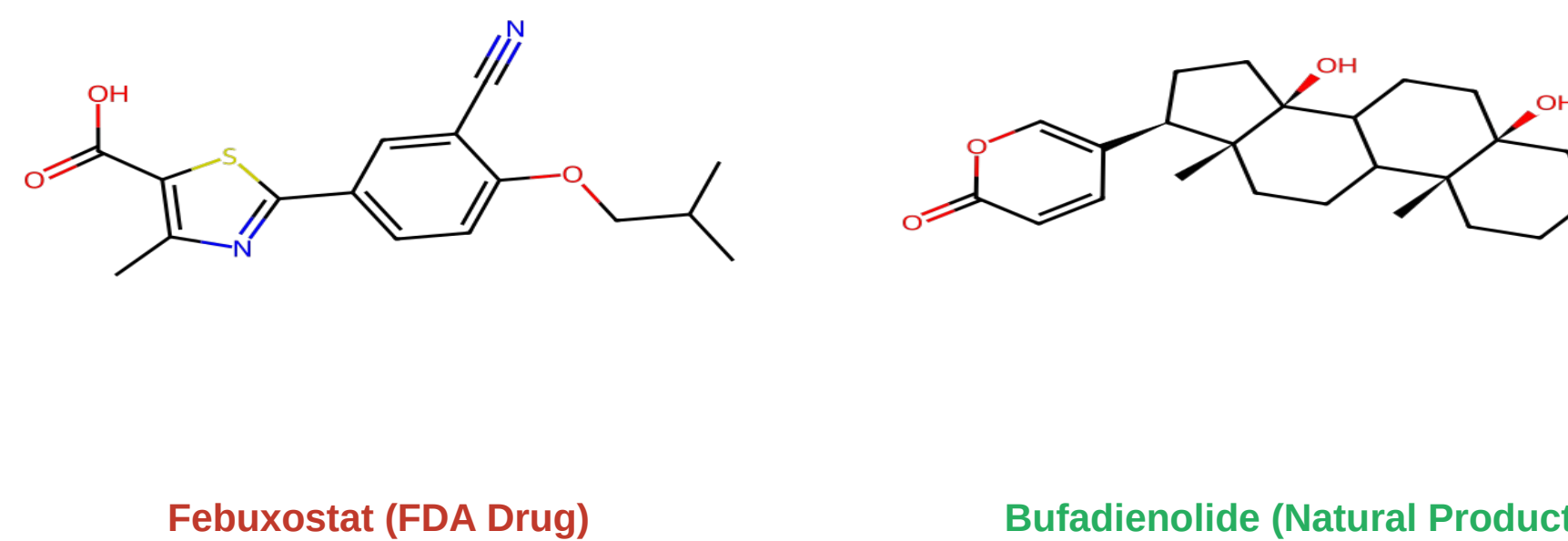
Key Finding: R702W mutation RIGIDIFIES the protein

- WT: Higher RMSD (4.05Å), more conformational sampling
- R702W: Lower RMSD (3.27Å), 0% frames >5Å
- Reduced flexibility may impair ligand recognition

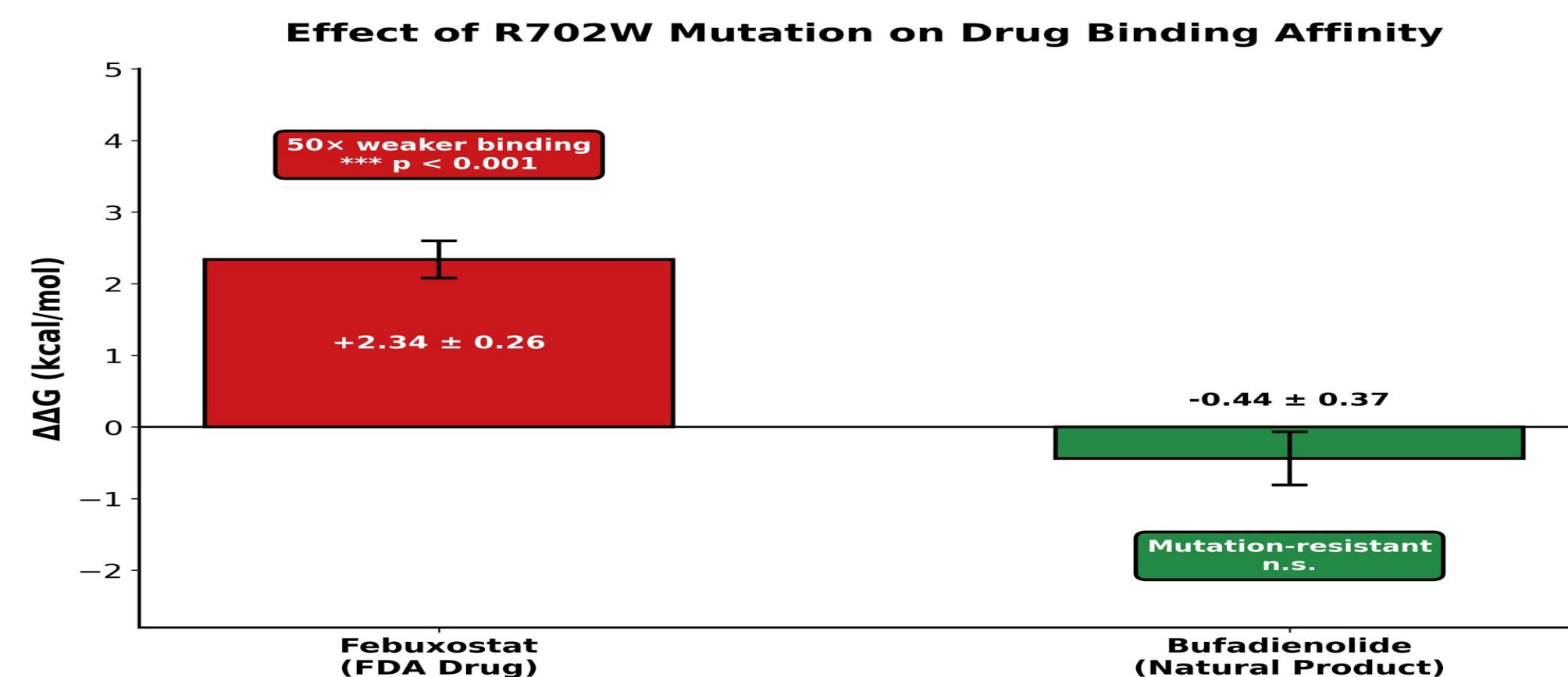
NOD2-SCOUT ML CLASSIFIER



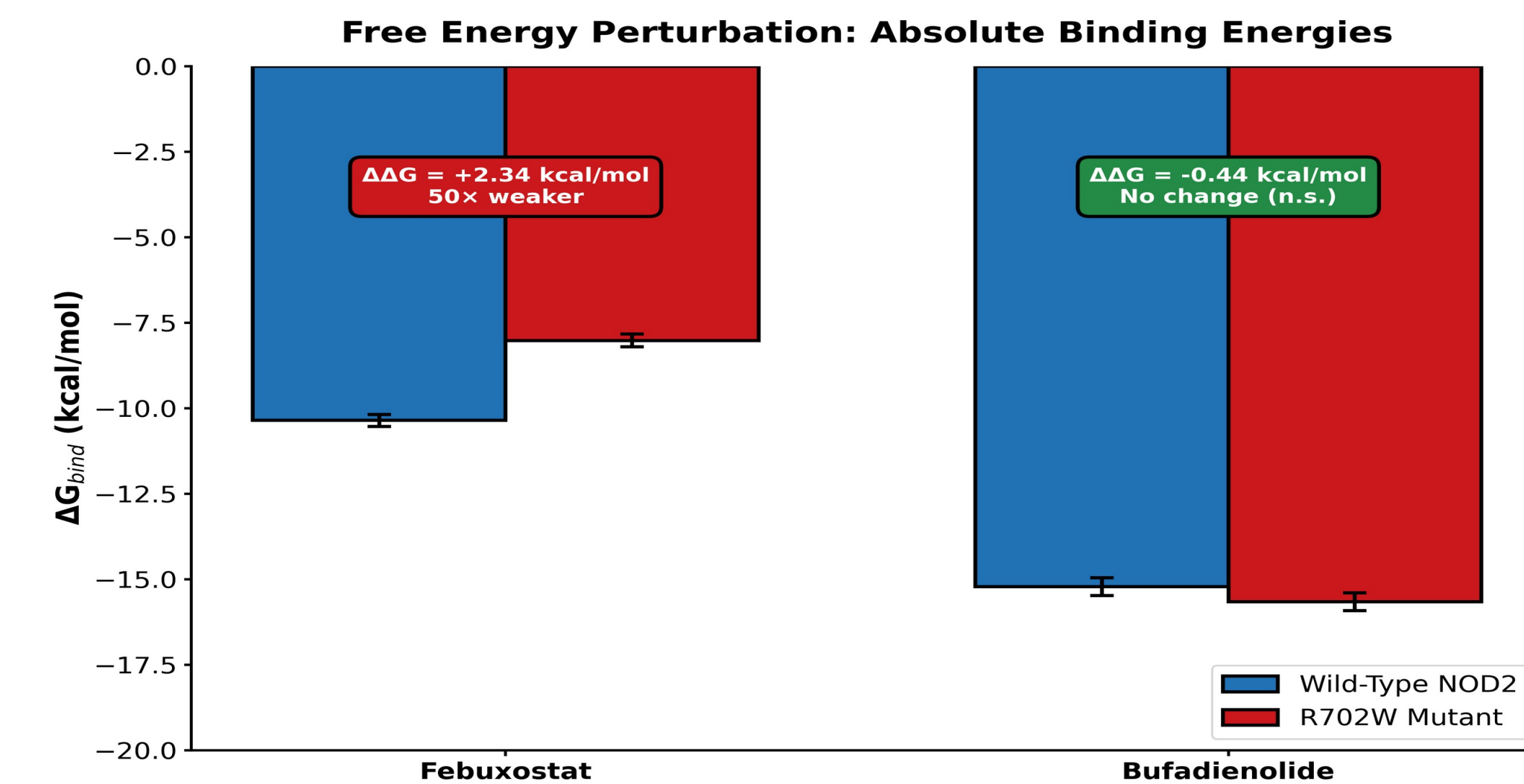
LEAD COMPOUNDS



FREE ENERGY PERTURBATION RESULTS



ABSOLUTE BINDING ENERGIES



KEY FINDINGS

- 1. R702W increases protein rigidity**
RMSD: 3.27Å vs 4.05Å ($p<0.05$)
- 2. Febuxostat: Mutation-SENSITIVE**
 $\Delta\Delta G = +2.34$ kcal/mol → 50× weaker in R702W
- 3. Bufadienolide: Mutation-RESISTANT**
 $\Delta\Delta G = -0.44$ kcal/mol → Binds equally
- 4. First reported NOD2-Bufadienolide interaction**

CONCLUSIONS

- ✓ Developed first computational pipeline for mutation-sensitive NOD2 drug discovery
- ✓ Identified Bufadienolide as novel mutation-resistant therapeutic lead
- ✓ Demonstrated allosteric mechanism linking distant mutation to binding site

FUTURE DIRECTIONS

- Experimental validation: SPR, ITC, HDX-MS
- Extend to G908R and 1007fs variants
- Structure-activity relationship studies
- Cell-based functional assays

ACKNOWLEDGMENTS

Computing: Personal RTX 4060 Ti, Google Colab Pro
Software: GNINA, GROMACS, pmx, AlphaFold2, RDKit